The bilayer dipolar-exciton interaction is completely monotone in r^2: kernel-checked triangular-lattice corollaries at every density and an exact-Ewald CVODE consistency study

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AI-assistance disclosure. This preprint, its Lean code, its Rust and Python code and its claim ledger were produced by AI agents (Claude, Anthropic) under the author's direction. The author is responsible for the content. No human has audited the Lean statements for adequacy, and no human referee has checked the hand proofs. A separate AI agent instance audited this preprint (verdict ACCEPT_WITH_CORRECTIONS, no blocking findings); this version applies its corrections (`AUDIT.md`, `REVISION.md`).

Abstract

Two rigid, vertically aligned electron–hole dipoles in a bilayer, at layer separation d and in-plane distance r, interact by $V(r) = 2\big(1/r - 1/\sqrt{r^2+d^2}\big)$ in units of $e^2/(4\pi\varepsilon)$. As a function of $t = r^2$ this is $g_d(t) = 2\big(t^{-1/2} - (t+d^2)^{-1/2}\big)$.

Tier A (Lean 4.34.1, pinned openai/math). The axiom footprint is {`propext`, `Classical.choice`, `Quot.sound`}, and the planted `sorry` and planted-axiom controls are caught. We prove that g_d is admissible in upstream's sense (smooth, nonnegative, completely monotone on $(0,\infty)$). The same holds for the whole family $t^{-s/2} - (t+c)^{-s/2}$ with $s > 0$, $c \geq 0$; for $c < 0$ it fails, a negative control. With upstream's `universal_energy_minimum` this gives triangular-lattice optimality among locally finite configurations at density one and, through a rescaling theorem, at every density $\rho > 0$. For g_d ($d > 0$) we also prove that the optimal energy is finite, at density one and at every density, so those inequalities are not $\top \leq \top$; no finiteness theorem is proved for the rest of the family. The algebraic identity $g_d(r^2) = 2(1/r - 1/\sqrt{r^2+d^2})$ is kernel-checked as well. Every optimality statement is *conditional* on that upstream theorem, which is AI-generated and not peer reviewed (the planar Cohn–Kumar conjecture was stated open in the human literature in 2019 [2]), and on the fidelity of its definitions.

Tier C. The physics reading and a step-by-step human proof for undergraduates. Complete monotonicity of g_d is elementary; the only new item is the explicit kernel-checked corollary. We claim no breakthrough.

Tier X. A new rusty-SUNDIALS module computes exact Ewald energies of the untruncated potential. It agrees with an independent Python Ewald sum to $\leq 1.9 \times 10^{-13}$ relative. CVODE gradient-flow relaxations found 0 of 21 relaxed periodic competitors below the triangular lattice; only 14 of them are open searches, the other 7 are designed negative controls. A HuggingFace JARVIS-DFT mirror supplies a proxy layer spacing $d \approx 6.599$ Å for $MoSe_2/WSe_2$, which places the experimental exciton densities at $\tilde{d} = d\sqrt{n} \approx 0.02$–$0.11$.

Tiers. [A]: kernel-checked Lean, with axioms within the allowlist and planted controls caught. [L]: a cited source, quoted in the ledger (prose as extracted, mathematics transcribed linearly, elisions marked). [C]: an argument or human proof that is not machine-checked. [X]: floating-point exploration, which never supports a claim. Identifiers such as `DX-A-0004` refer to `ledger/ledger.json` (Section 7). A Tier A tag attaches to the Lean statement, never to its physical gloss. Status words: the four prior-round theorems (`DipolarExciton.lean`) were recompiled by a separate AI verifier with verdict ACCEPT_WITH_CORRECTIONS; the new module (`DipolarExcitonMultipolar.lean`) was re-elaborated from source by a separate AI auditor at the stage-4 audit (rc 0, axioms within the allowlist, the auditor's own planted `sorry` and planted axiom caught); the small Step-1 module added in this revision (`DipolarExcitonStep1.lean`) is *producer-checked only.* No module has had a human audit.

1 Setting

Upstream [1] calls $g : \mathbb{R} \to \mathbb{R}$ *admissible* if it is C^∞ on $(0, \infty)$, $g \geq 0$ there, and $(-1)^n g^{(n)} \geq 0$ for all n. The pair interaction of two points x, y in the plane is $g(\|x - y\|^2)$: g is a function of the *squared* distance. The energy of a configuration C is the lim inf over radii R of the average, over points in the closed disk of radius R, of the sum over all other points in the disk; this is the lower energy (1.1) of Petrache–Serfaty [2] [L, DX-L-0001]. `latticeEnergy` sums $g(\|a\|^2)$ over all nonzero points of the density-one triangular lattice A. Upstream proves that A has the least energy among locally finite density-one configurations [A, DX-A-0001]; the rescaling theorem D2 of [6] extends this to every density [A, DX-A-0002]. Petrache and Serfaty (2019) state that the planar Cohn–Kumar conjecture "remains open in dimension 2" [L, DX-L-0001]; we did not re-check its status after 2019 for this paper, so DX-A-0001 is an AI-generated claim on an open problem, accepted here only as a kernel-checked statement about upstream's definitions.

Listing 1: Upstream definitions and theorems (openai/math pinned copy, and D2 `AnyDensity.lean`; proofs elided).

```
def AdmissiblePotential (g : ℝ → ℝ) : Prop :=
  ContDiffOn ℝ (⊤ : ℕ∞) g (Set.Ioi 0) ∧
  (∀ t : ℝ, 0 < t → 0 ≤ g t) ∧
  ∀ (r : ℕ) (t : ℝ), 0 < t → 0 ≤ (-1 : ℝ) ^ r * iteratedDeriv r g t

def diskEnergy (g : ℝ → ℝ) (C : Set Plane) (R : ℝ) : ℝ≥0∞ :=
  (diskCount C R : ℝ≥0∞)⁻¹ *
    ∑ x ∈ diskPoints C R, ∑ y ∈ (diskPoints C R).erase x,
      ENNReal.ofReal (g (‖x - y‖ ^ 2))

def energy (g : ℝ → ℝ) (C : Set Plane) : ℝ≥0∞ :=
  Filter.liminf (diskEnergy g C) atTop

def latticeEnergy (g : ℝ → ℝ) : ℝ≥0∞ :=
  ∑' a : {x : Plane // x ∈ triangularLattice ∧ x ≠ 0},
    ENNReal.ofReal (g (‖a.val‖ ^ 2))

theorem universal_energy_minimum (g : ℝ → ℝ) (C : Set Plane)
    (hg : AdmissiblePotential g) (_hC : LocallyFinite C) (hd : DensityOne C) :
    latticeEnergy g ≤ energy g C ∧ latticeEnergy g = energy g A := -- proof elided

def DensityRho (ρ : ℝ) (C : Set Plane) : Prop :=
  Tendsto (fun R : ℝ => (diskCount C R : ℝ) / (Real.pi * R ^ 2)) atTop (𝓝 ρ)

theorem triangular_optimal_any_density (h : ℝ → ℝ) (hh : AdmissiblePotential h) {ρ : ℝ} (hρ : 0
     < ρ) :
```

```
    LocallyFinite ((Real.sqrt ρ)⁻¹ • A) ∧ DensityRho ρ ((Real.sqrt ρ)⁻¹ • A) ∧
      ∀ C : Set Plane, LocallyFinite C → DensityRho ρ C → energy h ((Real.sqrt ρ)⁻¹ • A) ≤
    energy h C := -- proof elided
```

2 Step-by-step human demonstration

This section is for a strong undergraduate. Each step states the fact, proves it, and names the Lean declaration that checks it. The physics of Step 1 is [C] and its algebra [A]; Steps 2–9 are mirrored in Lean [A].

Listing 2: The potential, its negative control, and the shifted-difference family (verbatim).

```
noncomputable def g (d : ℝ) : ℝ → ℝ := fun t => 2 * (t ^ (-(1 / 2 : ℝ)) - (t + d ^ 2) ^ (-(1 /
    2 : ℝ)))

noncomputable def h (d : ℝ) : ℝ → ℝ := fun t => t ^ (-(1 / 2 : ℝ)) - 2 * (t + d ^ 2) ^ (-(1 / 2
     : ℝ))

noncomputable def shiftDiff (s c : ℝ) : ℝ → ℝ := fun t => t ^ (-(s / 2)) - (t + c) ^ (-(s / 2))
```

Step 1: the pair potential [C, DX-C-0001]. Exciton i is an electron at height d directly above a hole at height 0, both at in-plane position x_i. For two excitons with $|x_i - x_j| = r$ there are four Coulomb pairs: the two electrons (distance r, $+1/r$), the two holes ($+1/r$), and two electron–hole cross pairs at distance $\sqrt{r^2+d^2}$ (each $-1/\sqrt{r^2+d^2}$). The binding inside each exciton is a constant and is dropped. Hence $V(r) = 2/r - 2/\sqrt{r^2+d^2}$, and with $t = r^2$ we get $V(r) = g_d(r^2)$. Since $\sqrt{r^2+d^2} > r$, $V > 0$, and at large r, $V(r) = d^2/r^3 + O(r^{-5})$, the dipole–dipole law. The pair counting (which charges sit where, the factor 2, dropping the intra-exciton binding) is the physical reading and stays at Tier C. The algebra is kernel-checked [A, DX-A-0013]: $g_d(t) = 2(1/\sqrt{t} - 1/\sqrt{t+d^2})$ for $t > 0$ (`g_sqrt`), hence $g_d(r^2) = 2(1/r - 1/\sqrt{r^2+d^2})$ for $r > 0$ (`g_inplane`, using $\sqrt{r^2} = r$), and $g_d(r^2) > 0$ for $d \neq 0$ (`g_inplane_pos`); Listing 7. The proof of `g_sqrt` is the stage-4 auditor's probe, moved unchanged into a claimed module.

Step 2: derivatives of a shifted power. For real a, $c \geq 0$, $t > 0$ and $r \in \mathbb{N}$,

$$\frac{d^r}{dt^r}(t+c)^a = a(a-1)\cdots(a-r+1)\,(t+c)^{a-r} = \mathrm{desc}_r(a)\,(t+c)^{a-r}.$$

Proof. Induction on r: differentiating $(t+c)^{a-r}$ gives $(a-r)(t+c)^{a-r-1}$, and $\mathrm{desc}_{r+1}(a) = \mathrm{desc}_r(a)(a-r)$. The base $t + c > 0$, so every power is differentiable there. By linearity, $g_d^{(r)}(t) = 2\,\mathrm{desc}_r(-\frac{1}{2})\big(t^{-1/2-r} - (t+d^2)^{-1/2-r}\big)$. Lean: `iteratedDeriv_rpow`, `iteratedDeriv_shift`, `iteratedDeriv_g` (Listing 3).

Step 3: the sign. $(-1)^r \mathrm{desc}_r(-\frac{1}{2}) = \prod_{k=0}^{r-1}(-1)(-\frac{1}{2}-k) = \prod_{k=0}^{r-1}(\frac{1}{2}+k) = \mathrm{asc}_r(\frac{1}{2}) > 0$, a product of positive numbers. Lean: `sign_desc` (Mathlib's `ascPochhammer_eval_neg_eq_descPochhammer` and `ascPochhammer_pos`).

Step 4: the bracket. For $e \leq 0$ the map $x \mapsto x^e$ is nonincreasing on $(0,\infty)$. With $e = -\frac{1}{2} - r$ and $t \leq t + d^2$ we get $t^e - (t+d^2)^e \geq 0$. Lean: `diff_nonneg`.

Step 5: admissibility [A, DX-A-0003]. Steps 2–4 give $(-1)^r g_d^{(r)}(t) = 2\,\mathrm{asc}_r(\frac{1}{2})\big(t^{-1/2-r} - (t+d^2)^{-1/2-r}\big) \geq 0$; $r=0$ is $g_d \geq 0$. Smoothness: $t \mapsto t^a$ and $t \mapsto (t+c)^a$ are C^∞ where the base is positive. So g_d is admissible. Replacing $\frac{1}{2}$ by $s/2$ ($s>0$) and d^2 by $c \geq 0$ changes nothing, which gives `shiftDiff_admissible`; a nonnegative multiple of an admissible function is admissible (`admissible_const_mul`), and $g_d = 2\,\mathrm{shiftDiff}_{1,d^2}$ (`g_eq_shiftDiff`), so g_d is admissible for *every* real d (`dipolar_admissible_all`) [A, DX-A-0007]. Statements: Listings 4 and 5. (Equivalently, by Bernstein, $t^{-1/2} - (t+c)^{-1/2} = \pi^{-1/2}\int_0^\infty e^{-ut}(1-e^{-uc})u^{-1/2}\,du$, a positive mixture of exponentials [C, DX-C-0003].)

Step 6: triangular optimality at density one [A, DX-A-0004]. Apply upstream (Listing 1) to g_d: for every locally finite density-one C, $\mathrm{latticeEnergy}(g_d) \leq \mathrm{energy}(g_d, C)$. This step is exactly as strong as upstream. Lean: `triangular_dipolar_optimal`.

Step 7: the bound is not $\top \leq \top$ [A, DX-A-0005]. Put $a = t^{-1/2}$, $b = (t+c)^{-1/2}$, so $0 < b \leq a$, $a^2 t = 1$ and $b^2(t+c) = 1$. Then $a^2 - b^2 = a^2b^2\big((t+c)-t\big) = c\,a^2b^2$, i.e. $(a-b)(a+b) = c\,a^2b^2$. Since $b \leq a$, $2b^2 \leq a^2 + ab = a(a+b)$, hence $2(a-b)(a+b) \leq c\,a^3(a+b)$, and dividing by $a+b>0$, $a - b \leq \frac{c}{2}\,t^{-3/2}$. With $c = d^2$: $g_d(t) \leq d^2 t^{-3/2}$. The triangular lattice sum of $\|a\|^{-3}$ converges (Riesz exponent $3 > 2$), so $\mathrm{latticeEnergy}(g_d) \leq d^2\,\mathrm{latticeEnergy}(t^{-3/2}) < \infty$. Lean: `diff_le`, `g_le`, `latticeEnergy_dipolar_lt_top` (with `latticeEnergy_riesz_lt_top`).

Step 8: every density [A, DX-A-0010, DX-A-0011]. If C has density ρ, then $\sqrt{\rho}\,C$ has density one, and $\|\sqrt{\rho}x - \sqrt{\rho}y\|^2 = \rho\|x-y\|^2$, so the g-energy of C is the $g(\cdot/\rho)$-energy of $\sqrt{\rho}\,C$. The function $t \mapsto g(t/\rho)$ is admissible (each derivative gains the positive factor ρ^{-r}). Upstream applied to $g(\cdot/\rho)$ then says $\rho^{-1/2}A$ is optimal at density ρ. This is D2's `triangular_optimal_any_density`, instantiated as `triangular_dipolar_optimal_any_density`. Non-vacuity: $g_d(t/\rho) \leq d^2\rho^{3/2}t^{-3/2}$ (`g_scaled_le`), so the optimal energy is finite (`energy_dipolar_any_density_lt_top`); statements in Listing 6. Physically, $g_d(t/\rho) = \sqrt{\rho}\,g_{d\sqrt{\rho}}(t)$: density ρ at separation d is density one at separation $\tilde{d} = d\sqrt{\rho}$, with energies scaled by $\sqrt{\rho}$ [C, DX-C-0004]; this identity is not kernel-checked and is not needed by Lean.

Step 9: negative controls [A, DX-A-0006, DX-A-0008]. (i) $h_d(t) = t^{-1/2} - 2(t+d^2)^{-1/2}$ (cross attraction weighted twice): at $t = d^2$, $h_d = 1/d - 2(2d^2)^{-1/2} < 1/d - 2(4d^2)^{-1/2} = 0$, so h_d is not admissible (`h_neg_at`, `h_not_admissible`). (ii) For $c<0$ and $t = -2c > 0$, $t + c = -c \in (0,t)$, so $(t+c)^{-s/2} > t^{-s/2}$ and $\mathrm{shiftDiff}_{s,c}(t) < 0$ (`shiftDiff_not_admissible_of_neg`). The sign of the shift is load-bearing, and a gate that accepted these would be broken.

Remark 1 (scope of non-vacuity). The finiteness theorems of Steps 7–8 are proved for g_d only. The family corollaries `triangular_shiftDiff_optimal` and `triangular_shiftDiff_optimal_any_density` [A, DX-A-0009] carry no finiteness theorem; at $c=0$, $\mathrm{shiftDiff}_{s,0} \equiv 0$ and their conclusion is the trivial $0 \leq$ energy.

Listing 3: Step lemmas (statements verbatim, proofs elided).

```
lemma iteratedDeriv_rpow (a : ℝ) (r : ℕ) (t : ℝ) :
    iteratedDeriv r (fun x : ℝ => x ^ a) t = (descPochhammer ℝ r).eval a * t ^ (a - r) := --
    proof elided

lemma iteratedDeriv_shift (a c : ℝ) (r : ℕ) (t : ℝ) :
    iteratedDeriv r (fun x : ℝ => (x + c) ^ a) t = (descPochhammer ℝ r).eval a * (t + c) ^ (a - r
    ) := -- proof elided
```

```
lemma iteratedDeriv_g (d : ℝ) (r : ℕ) {t : ℝ} (ht : 0 < t) :
    iteratedDeriv r (g d) t =
      2 * ((descPochhammer ℝ r).eval (-(1 / 2 : ℝ)) *
        (t ^ (-(1 / 2 : ℝ) - r) - (t + d ^ 2) ^ (-(1 / 2 : ℝ) - r))) := -- proof elided

lemma sign_desc (r : ℕ) :
    (-1 : ℝ) ^ r * (descPochhammer ℝ r).eval (-(1 / 2 : ℝ)) = (ascPochhammer ℝ r).eval (1 / 2 :
    ℝ) := -- proof elided

lemma diff_nonneg (c e : ℝ) (hc : 0 ≤ c) (he : e ≤ 0) {t : ℝ} (ht : 0 < t) :
    0 ≤ t ^ e - (t + c) ^ e := -- proof elided
```

3 Lean formalization

Prior-round module `formal_cert/Autoresearch/DipolarExciton.lean` (namespace `AutoresearchDipolar`), imported unchanged, and the new module `DipolarExcitonMultipolar.lean` (namespace `AutoresearchDipolarMultipolar`). Both are compiled with the pinned toolchain (Lean 4.34.1, Mathlib `d13f23b7`, openai/math `fd4aeeb2`). `grep -c sorry` on both claimed modules gives 0. Planted controls live only in `DipolarExcitonMultipolarControls.lean`. At the stage-4 audit a separate AI agent recompiled both modules from source (`DipolarExciton.lean` rc 0 in 62.9 s, `DipolarExcitonMultipolar.lean` rc 0 in 56.9 s; `audit/recompile_*_src.json`) and caught its own planted `sorry` and planted axiom in a separate probe file (`audit/audit_controls.json`); all 16 re-printed claimed declarations stayed within the allowlist. This revision adds a third claimed module, `DipolarExcitonStep1.lean` (namespace `AutoresearchDipolarStep1`, rc 0 in 22.7 s; `grep -c sorry` gives 0), with planted controls in `DipolarExcitonStep1Controls.lean` (rc 0 in 16.4 s): `ctl1_sorry` is flagged with `sorryAx`, `ctl1_axiom` with the planted `ctl1_cheat`, and `ctl1_wrong_sign_false` proves that the wrong-sign in-plane form fails at $r = d = 1$ [A, DX-A-0014]:

```
rc = 0, 16.4 s
formal_cert/Autoresearch/DipolarExcitonStep1Controls.lean:9:8: warning: declaration uses `sorry`
'AutoresearchDipolarStep1Controls.ctl1_sorry' depends on axioms: [sorryAx]
'AutoresearchDipolarStep1Controls.ctl1_axiom' depends on axioms: [AutoresearchDipolarStep1Controls.
    ctl1_cheat]
'AutoresearchDipolarStep1Controls.ctl1_wrong_sign_false' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'AutoresearchDipolarStep1.g_sqrt' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchDipolarStep1.g_inplane' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchDipolarStep1.g_inplane_pos' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Listing 4: Prior-round theorems (verbatim statements).

```
theorem dipolar_admissible {d : ℝ} (_hd : 0 < d) : AdmissiblePotential (g d) := -- proof elided

theorem triangular_dipolar_optimal {d : ℝ} (hd : 0 < d) (C : Set Plane) (hC : LocallyFinite C)
    (hdens : DensityOne C) : latticeEnergy (g d) ≤ energy (g d) C := -- proof elided

theorem h_neg_at {d : ℝ} (hd : 0 < d) : h d (d ^ 2) < 0 := -- proof elided

theorem h_not_admissible {d : ℝ} (hd : 0 < d) : ¬ AdmissiblePotential (h d) := -- proof elided

lemma diff_le (c : ℝ) (hc : 0 ≤ c) {t : ℝ} (ht : 0 < t) :
    t ^ (-(1 / 2 : ℝ)) - (t + c) ^ (-(1 / 2 : ℝ)) ≤ c / 2 * t ^ (-(3 / 2 : ℝ)) := -- proof
    elided
```

```
lemma g_le (d : ℝ) {t : ℝ} (ht : 0 < t) : g d t ≤ d ^ 2 * riesz 3 t := -- proof elided

theorem latticeEnergy_dipolar_lt_top (d : ℝ) : latticeEnergy (g d) < ⊤ := -- proof elided
```

Listing 5: New: the shifted-difference family and its negative control (independently recompiled by an AI auditor; no human audit).

```
theorem shiftDiff_admissible {s c : ℝ} (hs : 0 < s) (hc : 0 ≤ c) : AdmissiblePotential (
    shiftDiff s c) := -- proof elided

lemma admissible_const_mul {f : ℝ → ℝ} (hf : AdmissiblePotential f) {k : ℝ} (hk : 0 ≤ k) :
    AdmissiblePotential (fun t => k * f t) := -- proof elided

lemma g_eq_shiftDiff (d : ℝ) : AutoresearchDipolar.g d = fun t => 2 * shiftDiff 1 (d ^ 2) t := --
     proof elided

theorem dipolar_admissible_all (d : ℝ) : AdmissiblePotential (AutoresearchDipolar.g d) := --
    proof elided

theorem shiftDiff_neg_of_neg {s c : ℝ} (hs : 0 < s) (hc : c < 0) : shiftDiff s c (-2 * c) < 0 :=
    -- proof elided

theorem shiftDiff_not_admissible_of_neg {s c : ℝ} (hs : 0 < s) (hc : c < 0) :
    ¬ AdmissiblePotential (shiftDiff s c) := -- proof elided

theorem triangular_shiftDiff_optimal {s c : ℝ} (hs : 0 < s) (hc : 0 ≤ c) (C : Set Plane) (hC :
    LocallyFinite C)
    (hdens : DensityOne C) : latticeEnergy (shiftDiff s c) ≤ energy (shiftDiff s c) C := -- proof
     elided
```

Listing 6: New: every density and non-vacuity for g_d (independently recompiled by an AI auditor; no human audit).

```
theorem triangular_dipolar_optimal_any_density {d : ℝ} (hd : 0 < d) {ρ : ℝ} (hρ : 0 < ρ) :
    LocallyFinite ((Real.sqrt ρ)⁻¹ • A) ∧ TriangularDensity.DensityRho ρ ((Real.sqrt ρ)⁻¹ • A) ∧
      ∀ C : Set Plane, LocallyFinite C → TriangularDensity.DensityRho ρ C →
        energy (AutoresearchDipolar.g d) ((Real.sqrt ρ)⁻¹ • A) ≤ energy (AutoresearchDipolar.g d)
     C := -- proof elided

lemma g_scaled_le (d : ℝ) {ρ : ℝ} (hρ : 0 < ρ) {t : ℝ} (ht : 0 < t) :
    AutoresearchDipolar.g d (ρ⁻¹ * t) ≤ (d ^ 2 * (ρ⁻¹) ^ (-((3 : ℝ) / 2))) * riesz 3 t := --
    proof elided

theorem latticeEnergy_dipolar_scaled_lt_top (d : ℝ) {ρ : ℝ} (hρ : 0 < ρ) :
    latticeEnergy (fun t => AutoresearchDipolar.g d (ρ⁻¹ * t)) < ⊤ := -- proof elided

theorem energy_dipolar_any_density_lt_top {d : ℝ} (hd : 0 < d) {ρ : ℝ} (hρ : 0 < ρ) :
    energy (AutoresearchDipolar.g d) ((Real.sqrt ρ)⁻¹ • A) < ⊤ := -- proof elided
```

Listing 7: Revision: the Step-1 identity (producer-checked only).

```
theorem g_sqrt (d t : ℝ) (ht : 0 < t) :
    AutoresearchDipolar.g d t = 2 * (1 / Real.sqrt t - 1 / Real.sqrt (t + d ^ 2)) := -- proof
    elided

theorem g_inplane (d r : ℝ) (hr : 0 < r) :
```

```
    AutoresearchDipolar.g d (r ^ 2) = 2 * (1 / r - 1 / Real.sqrt (r ^ 2 + d ^ 2)) := -- proof
    elided

theorem g_inplane_pos (d r : ℝ) (hd : d ≠ 0) (hr : 0 < r) : 0 < AutoresearchDipolar.g d (r ^ 2)
    := -- proof elided
```

Axiom audit and planted controls [A, DX-A-0012]. The controls file imports the built module, plants `theorem ctl_sorry : (1 : ℕ) = 2 := sorry` and `axiom ctl_cheat : False`, and prints the axioms of every claimed theorem. Verbatim output (stage 1, `contribution/compile_controls.json`):

```
rc = 0, 29.7 s
formal_cert/Autoresearch/DipolarExcitonMultipolarControls.lean:8:8: warning: declaration uses `sorry`
'AutoresearchDipolarMultipolarControls.ctl_sorry' depends on axioms: [sorryAx]
'AutoresearchDipolarMultipolarControls.ctl_axiom' depends on axioms: [AutoresearchDipolarMultipolarControls.
    ctl_cheat]
'AutoresearchDipolar.dipolar_admissible' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchDipolar.triangular_dipolar_optimal' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchDipolar.latticeEnergy_dipolar_lt_top' depends on axioms: [propext, Classical.choice, Quot.sound
    ]
'AutoresearchDipolar.h_not_admissible' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchDipolarMultipolar.shiftDiff_admissible' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'AutoresearchDipolarMultipolar.admissible_const_mul' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'AutoresearchDipolarMultipolar.g_eq_shiftDiff' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchDipolarMultipolar.dipolar_admissible_all' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'AutoresearchDipolarMultipolar.shiftDiff_neg_of_neg' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'AutoresearchDipolarMultipolar.shiftDiff_not_admissible_of_neg' depends on axioms: [propext,
 Classical.choice,
 Quot.sound]
'AutoresearchDipolarMultipolar.triangular_dipolar_optimal_any_density' depends on axioms: [propext,
 Classical.choice,
 Quot.sound]
'AutoresearchDipolarMultipolar.triangular_shiftDiff_optimal_any_density' depends on axioms: [propext,
 Classical.choice,
 Quot.sound]
'AutoresearchDipolarMultipolar.triangular_shiftDiff_optimal' depends on axioms: [propext, Classical.choice,
    Quot.sound]
'AutoresearchDipolarMultipolar.g_scaled_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'AutoresearchDipolarMultipolar.latticeEnergy_dipolar_scaled_lt_top' depends on axioms: [propext,
 Classical.choice,
 Quot.sound]
'AutoresearchDipolarMultipolar.energy_dipolar_any_density_lt_top' depends on axioms: [propext,
 Classical.choice,
 Quot.sound]
'TriangularDensity.triangular_optimal_any_density' depends on axioms: [propext, Classical.choice, Quot.sound
    ]
'OAI.AtomicTriangular.universal_energy_minimum' depends on axioms: [propext, Classical.choice, Quot.sound]
```

The file exits with rc 0 *despite* the planted `sorry` and axiom, so the exit code is not the gate; the axiom list is. Both controls are flagged (`sorryAx`, `ctl_cheat`); every claimed theorem is within the allowlist.

4 Physics: what the theorem does and does not say

System and units. Interlayer (indirect) excitons in a transition-metal-dichalcogenide heterobilayer such as $MoSe_2/WSe_2$: electrons in one layer, holes in the other, bound into dipoles of length $\approx d$.

Energies are in units of $e^2/(4\pi\varepsilon_0\varepsilon_r)$ per length; lengths in the numerics are in units of $n^{-1/2}$ (density one), and $\tilde{d} = d\sqrt{n}$ [C, DX-C-0004].

What it says [C, DX-C-0002]. In the classical, zero-temperature model of rigid point dipoles perpendicular to the layers, with unscreened Coulomb interactions, no locally finite configuration at any fixed density has lower lower-energy than the triangular lattice, conditional on upstream and on definition fidelity.

What it does not say. It is not a uniqueness statement (finite perturbations of A have the same lim inf energy [2]). It contains no kinetic or zero-point energy, so it says nothing about quantum (bosonic) exciton liquids or condensates. It has no dielectric screening (h-BN, Rytova–Keldysh), no moiré potential, no in-plane electron–hole offset, no separate electron and hole crystals, and no electron–hole plasma. Experiments on h-BN-encapsulated $MoSe_2/WSe_2$ at 4 K report moiré-trapped excitons below $n \sim 1$–5×10^{11} cm^{-2}, a free exciton gas only "in the narrow density range around $10^{11} \sim 10^{12}$ cm^{-2}", a Mott transition at $n_{\mathrm{Mott}} = 1.6$–3×10^{12} cm^{-2}, and moiré wells "of the order of 10s-100 meV" deep [3] [L, DX-L-0002]. In that window real interlayer excitons are a quantum gas, not a classical crystal [C, DX-C-0007].

5 rusty-SUNDIALS numerics

Everything in this section is [X]: it illustrates the Tier A statement and checks consistency, and supports no claim. Code: `crates/qf-yukawa-crystal` (`dipolar.rs`, `dipolar_ewald.rs`, `bin/dipolar_crystal.rs`) on branch `physics/ar-dipolar-exciton-2026-10-25`; the preregistration (DIP-TRI-1) was committed before the solver code (Section 9).

Method. The code uses the rescaled form $U = V/\tilde{d}^2 = 2/(r\,s\,(r+s))$, $s = \sqrt{r^2+\tilde{d}^2}$, which has no cancellation at small $\tilde{d}$ and tends to $1/r^3$. Exact energies of periodic configurations of the *untruncated* potential use an Ewald split of $V(r) = \frac{4}{\sqrt{\pi}} \int_0^\infty e^{-r^2u^2}(1 - e^{-\tilde{d}^2u^2})\,du$ at $u = \eta$: real-space part $2(\mathrm{erfc}(\eta r)/r - \mathrm{erfc}(\eta s)/s)$, reciprocal part including $G = 0$ (the potential is integrable, so no background is needed), and self term. The per-particle energy is $e = \frac{1}{2}\sum' U$, so latticeEnergy$(g_d) = 2d^2 e_{\mathrm{tri}}$ in these units [C, DX-C-0006]. Dynamics use a shifted-force truncation at $r_c = 8$ only to *generate* competitors: overdamped gradient flow integrated with CVODE (BDF, analytic Jacobian $= -$Hessian, rtol 10^{-10}, atol 10^{-12}) until $\max|F| < 10^{-9}$, for $N = 288$ particles in a $16a \times 9\sqrt{3}a$ box that holds the triangular lattice exactly. Every reported energy and gap is the Ewald value. *Deviation from DIP-TRI-1:* the preregistered `reported_potential` was the periodic image sum to radius R plus the analytic uniform tail; we report Ewald energies instead, because that truncated sum leaves the non-monotone residual described below (Figure 1).

Correctness controls [X, DX-X-0001]. An independent Python Ewald sum (adaptive `scipy` quadrature, $\eta = 1.0$ against 0.8 in Rust) agrees to $\le 8.5 \times 10^{-14}$ (triangular) and $\le 1.9 \times 10^{-13}$ (square) relative at the five $\tilde{d}$ values. Python direct sums with $R = 100$–400 plus the exact uniform tail $2\pi n/(\sqrt{R^2+\tilde{d}^2} + R)$ agree to $\le 1.9 \times 10^{-5}$, the size of the oscillating lattice-minus-continuum residual. *Negative control:* dropping the tail gives an error of 3.14×10^{-2}. The Rust energy is η-independent to $\le 1.3 \times 10^{-13}$ (five $\tilde{d}$, $\eta \in \{0.6, 0.8, 1.2\}$) and $\le 4.8 \times 10^{-13}$ (41-point grid), and the 288-particle box matches the primitive cell to $\le 2.6 \times 10^{-14}$. As $\tilde{d} \to 0$, $e_{\mathrm{tri}} \to 4.4462$ at $\tilde{d} = 0.01$, which should be half the density-one triangular Riesz-3 lattice sum; we did not compare it with

Table 1: Exact (Ewald) per-particle energies in $U = V/\tilde{d}^2$ units and lattice gaps [X, DX-X-0002].

$\tilde{d}$	e_{tri}	$e_{\mathrm{sq}} - e_{\mathrm{tri}}$	$e_{\mathrm{hc}} - e_{\mathrm{tri}}$	$e_{\mathrm{sq}}/e_{\mathrm{tri}} - 1$
0.0209	4.445599702966	7.0378e−02	4.2354e−01	1.583e−02
0.066	4.438685393575	6.9837e−02	4.2033e−01	1.573e−02
0.114	4.423567867904	6.8667e−02	4.1339e−01	1.552e−02
0.3	4.295977545078	5.9459e−02	3.5897e−01	1.384e−02
1	3.355596439147	1.9453e−02	1.2225e−01	5.797e−03

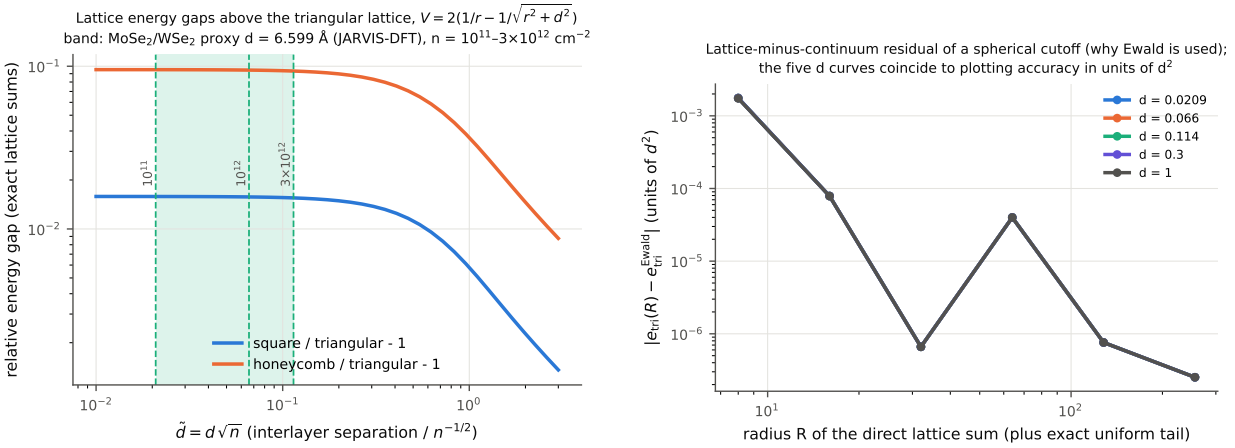


Figure 1: Left: exact square and honeycomb gaps relative to triangular against $\tilde{d}$ [X, DX-X-0002]. Right: residual of truncated sums with exact tail against R [X, DX-X-0005]. Generated by `scripts/dipolar_figures.py`.

a literature value. The stage-4 auditor's independent check (`audit/audit_epstein_check.py`, no Ewald sum: expansion in $\tilde{d}^2/r^2$ and closed-form Epstein zeta functions, mpmath, with its own positive and negative controls) reproduces e_{tri} and e_{sq} at $\tilde{d} = 0.0209, 0.066, 0.114, 0.3$ to about 10^{-13} relative and gives the exact $\tilde{d} \to 0$ limit 4.446373, consistent with the grid value [X, DX-X-0001]. A plain spherical cutoff with tail leaves a non-monotone residual of up to 1.7×10^{-3} at $R = 8$ and 4.0×10^{-5} at $R = 64$ (Figure 1) [X, DX-X-0005], which is why only Ewald gaps are reported.

Lattice gaps [X, DX-X-0002]. Table 1. On the 41-point grid $\tilde{d} \in [0.01, 3]$, $e_{\mathrm{sq}}/e_{\mathrm{tri}} - 1$ falls from 1.58×10^{-2} to 1.36×10^{-3} and the honeycomb gap from 9.53×10^{-2} to 8.75×10^{-3}; both stay positive. In the physical window the gaps are essentially the pure-dipolar values. These compare two lattices only.

Controls of the relaxation pipeline [X, DX-X-0003]. At all five $\tilde{d}$ (0.0209, 0.066, 0.114, 0.3, 1): the perfect and the noisy triangular lattices relax back (10 positive controls, $|\text{rel. gap}| \leq 2.6 \times 10^{-14}$, 0 defects, 0 negative Hessian modes). *Negative controls:* the square lattice is a saddle of the dynamics potential (64–72 negative Hessian modes) and square+noise relaxes to a distorted state still above triangular (rel. gap 1.73×10^{-3}–4.77×10^{-3}); two incommensurate boxes stay defected.

Main study, preregistered P1 [X, DX-X-0004]. 21 relaxed competitors: 12 random-start $T = 0$ quenches (at $\tilde{d} = 0.0209$, 0.066, 1), 2 anneals and 2 incommensurate boxes (at $\tilde{d} = 0.066$), and 5

Table 2: Relaxed competitors (exact relative gap to the triangular lattice; Hessian of the dynamics potential, – = not computed) [X, DX-X-0003, DX-X-0004].

$\tilde{d}$	run	family	rel. gap	defects	neg. modes
0.0209	ctl_neg_square_saddle	square+noise (neg.)	4.8×10^{-3}	0	–
0.0209	uniform_00	quench	1.4×10^{-3}	4	0
0.0209	uniform_01	quench	2.8×10^{-14}	0	0
0.066	ctl_neg_square_saddle	square+noise (neg.)	4.7×10^{-3}	0	–
0.066	incommensurate_00	incomm. box (neg.)	2.0×10^{-3}	8	–
0.066	incommensurate_01	incomm. box (neg.)	7.9×10^{-4}	4	–
0.066	uniform_00	quench	1.4×10^{-3}	4	0
0.066	uniform_01	quench	-2.0×10^{-16}	0	0
0.066	uniform_02	quench	2.5×10^{-3}	12	0
0.066	uniform_03	quench	3.5×10^{-3}	24	0
0.066	uniform_04	quench	1.5×10^{-4}	0	0
0.066	uniform_05	quench	1.4×10^{-3}	4	0
0.066	uniform_06	quench	1.2×10^{-3}	4	0
0.066	uniform_07	quench	7.9×10^{-4}	0	0
0.066	anneal_00	anneal	-5.2×10^{-15}	0	0
0.066	anneal_01	anneal	1.5×10^{-4}	0	0
0.114	ctl_neg_square_saddle	square+noise (neg.)	4.7×10^{-3}	0	–
0.3	ctl_neg_square_saddle	square+noise (neg.)	4.2×10^{-3}	0	–
1	ctl_neg_square_saddle	square+noise (neg.)	1.7×10^{-3}	0	–
1	uniform_00	quench	7.4×10^{-4}	8	0
1	uniform_01	quench	-2.3×10^{-14}	0	0

square+noise relaxations. Only the 14 quenches and anneals are open searches; the 2 incommensurate boxes and 5 square+noise runs (7 in all) are designed negative controls that are expected to sit above triangular. 0 lie below the exact triangular energy by more than 10^{-9} relative. The most negative competitor gap is -2.3×10^{-14} (rounding at the perfect lattice; -2.6×10^{-14} including the positive controls); the smallest gap among the 17 states not at the lattice is 1.54×10^{-4}. 3 of 12 quenches and 1 of 2 anneals reach the perfect lattice; 7 quenches end defected. Truncated and exact gaps differ by $\leq 1.6 \times 10^{-5}$. Caveats: seeds are reused across $\tilde{d}$, so the three lattice-reaching quenches share one start (seed 501); only $\tilde{d} = 0.066$ received the full preregistered run set; $\tilde{d} = 0.114$ and 0.3 have controls only; two zero-defect states sit above the lattice (probably strained crystallites, not analysed); two of the runs at $\tilde{d} = 0.066$, `anneal_01` and `uniform_04`, start from different seeds but end in the same state ($e = 4.43936745573261$, $\psi_6 = 0.998112$), so there are fewer independent outcomes than runs. The first attempts at $\tilde{d} = 0.0209$ and $\tilde{d} = 1$ (4 quenches, 1 anneal and 2 incommensurate boxes each, one thread) were killed after about 34 min because they could not finish within the budget; the driver then kept finished jobs only in memory, so none of their results was ever printed, and those two $\tilde{d}$ were rerun with 2 quenches each using the patched driver (commit `916fc88`) that prints every result. Table 2, Figure 2.

6 HuggingFace dataset illustration

Search. 26 queries of the Hub (`HfApi().list_datasets(search=...)`: C2DB, BiDB, bilayer, heterobilayer, JARVIS, TMD, MoS2, WSe2, interlayer, ...) found no dataset of TMD heterobilayers or of interlayer separations [X, DX-X-0006].

Dataset used. `colabfit/JARVIS_DFT_3D_12_12_2022`, revision `f4de0085edcf92885870c67459363d2c7080c90b`, file `co/co_0.parquet` (54,027,813 bytes), a ColabFit mirror of JARVIS-DFT [4, 5]. The card license is "unknown" [L, DX-L-0004]; only derived numbers are reproduced here, and the upstream figshare license was not checked. JARVIS-DFT uses

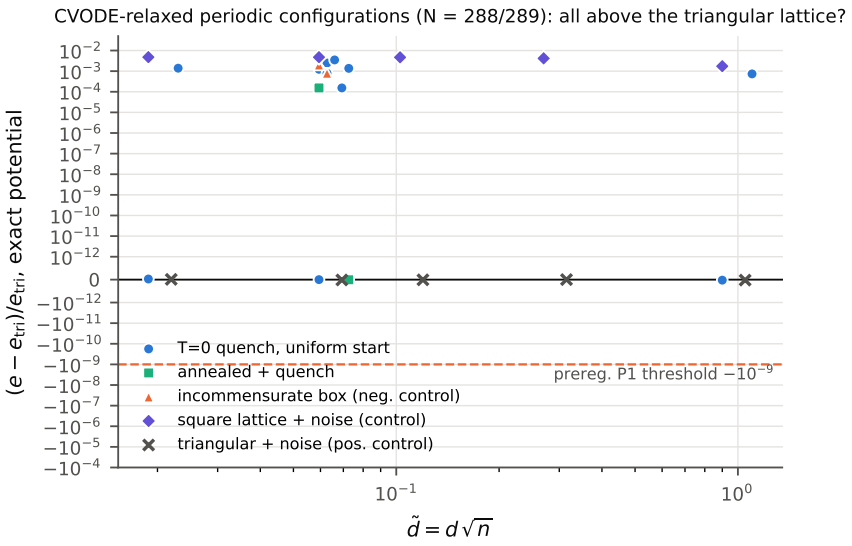


Figure 2: Exact relative gaps of relaxed states [X, DX-X-0004]. Generated by `scripts/dipolar_figures.py`.

Table 3: Adjacent metal-plane spacings from the HuggingFace JARVIS-DFT mirror (ColabFit names) [X, DX-X-0006].

material	rows	spacings	mean (Å)	range (Å)
$MoSe_2$	`jdft_3d-12-12-2022_27269`, `jdft_3d-12-12-2022_15445`, `jdft_3d-12-12-2022_24840`	6	6.5827	6.5818–6.5842
WSe_2	`jdft_3d-12-12-2022_35821`	3	6.6151	6.6131–6.6162
MoS_2	`jdft_3d-12-12-2022_33239`	4	6.2371	6.2361–6.2394

the vdW-DF-OptB88 functional, which its authors report gives accurate lattice parameters for van der Waals materials [4] [L, DX-L-0003].

What it shows [X, DX-X-0006]. A selection rule fixed before seeing the output (semiconducting rows, at least two MX_2 layers, metal-plane spacings 4.5–8 Å, vacuum gaps excluded) gives the adjacent metal-plane spacings of Table 3. The $MoSe_2/WSe_2$ proxy $d = \frac{1}{2}(6.5827 + 6.6151) = 6.599$ Å gives $\tilde{d} = 0.0209,\ 0.0660,\ 0.1143$ at $n = 10^{11}, 10^{12}, 3\times10^{12}$ cm^{-2}, the density window of [3]. The unscreened square-minus-triangular gap per exciton is then 0.0014, 0.044 and 0.223 meV (divide by ε_r) [X, DX-X-0007], about 1.5 to 5 orders of magnitude below the moiré depths of 10s–100 meV. These values multiply the physical $\tilde{d}^2$ by the gap $e_{\rm sq} - e_{\rm tri}$ evaluated at the nearest simulated $\tilde{d}$ (0.0209, 0.066, 0.114), not at the physical $\tilde{d}$ (0.02087, 0.06599, 0.1143); evaluating the gap at $\tilde{d} = 0.1143$, the stage-4 auditor gets 0.224 meV at 3×10^{12} cm^{-2}, a 0.5% difference.

Caveats. No heterobilayer row exists on the Hub, so the proxy comes from homo-stacks; the metal-plane spacing is used as the electron–hole separation, which is an assumption [C, DX-C-0005]; the MoS_2 row is a bulk-like cell, not a slab; no WS_2 row passed the selection. The dataset illustrates the physical $\tilde{d}$ range; it does not test the theorem.

7 Claim ledger

`ledger/build_ledger.py` files 33 claims: 14 at Tier A, 4 at L, 7 at C and 8 at X. Each has a content-addressed evidence blob in `ledger/evidence/`: verbatim axiom lines and source hashes (A), verbatim excerpts in `ledger/sources/excerpts.json` (L), argument text (C), and result files with `claim_licence: false` (X). The Elenchus gate (`tools/ledger.py ledger.json --evidence-dir evidence`, output in `ledger/gate_output.txt`) exits with code 1. It reports **0 blocking findings** and **14 flags**, all `LEDGER_UNAUDITED_TIER_A`: no person has certified that the Tier A statements mean what their glosses say. The flags remain open. The L excerpts are prose, not theorem statements, so the Elenchus citation gate (`citation_check.py`, theorem/lemma excerpts only) was not run on them.

8 Limitations

- **Upstream dependence.** Every triangular-lattice statement rests on openai/math's `universal_energy_minimum` (AI-generated, not peer reviewed; the human-literature conjecture was stated open in 2019 [2]) and on the fidelity of its definitions (`energy`, `LocallyFinite`, `DensityOne`, D2's `DensityRho`).
- **Verification status.** The prior-round module was re-checked by an AI verifier and the Multipolar module was re-elaborated from source by a separate AI auditor; neither by a person. The Step-1 module added in this revision is producer-checked only. The paper-stage reruns (Section 9) come from the same AI pipeline as the producer. The auditor did not re-run the crate tests (its isolation guard refused `cargo` on this worktree).
- **Novelty is small.** Complete monotonicity of g_d is elementary (Bernstein); the corollary is immediate given upstream. A literature search found no explicit statement, which is weak evidence [C, DX-C-0003].
- **Model.** Classical, rigid, unscreened point dipoles, no quantum kinetic energy, no moiré potential; real excitons in the physical window are a quantum gas.
- **Numerics.** 21 competitors from one box shape and N, of which only 14 are open searches and two share a final state; reduced run counts versus the preregistration; reused seeds; Ewald energies instead of the preregistered truncated sum. Tier X never supports the theorem.
- **Data.** Homo-stack proxy for a heterobilayer quantity; license "unknown" on the dataset card.

9 Reproducibility

AutoevolveAI files are uncommitted on top of base `db794dd98d12c2a79702eab5651ae7c36a35ea27` (branch `worktree-openai-math-discovery`). Lean source sha256 values (they match those recorded at stage 1):

- `DipolarExciton.lean`: `b85389a50373adaf877e866c79f0696ba0a284d9626bc9a56040821c3e11f27d`
- `DipolarExcitonMultipolar.lean`: `7f52ea64903c43cbfe95b76eea74a2ea01040c45d778ae05bbb848aa9554f98a`
- `DipolarExcitonMultipolarControls.lean`: `1754e1a9ba6ed844a1d0356872cf2a5662c89d13e85d1fd39747f3fe1215106d`
- `Triangular/AnyDensity.lean`: `0a6d86eef3ee24b2b6828ff602707868e160969bfbebc41db375c05b383c5168`

- `DipolarExcitonStep1.lean`: `ceb2c607498086e8f865ebeee944842548345501d704a54b0dfc465cd9f42f7d`
- `DipolarExcitonStep1Controls.lean`: `8a26b6ba14882ad6fc3e57cc69376786c30993695fc50846fc8a75add1ca4987`

(the last two, added in this revision, match the values recorded when they were compiled). rusty-SUNDIALS commits on branch `physics/ar-dipolar-exciton-2026-10-25` (not pushed):

- preregistration `4feb027eb502ff0936a28f8168499b6fef25e276`;
- code `17ea6beccd044a02f46a8d8f7f04c76f31ee33f0`;
- per-job output patch `916fc88c8cebab15426427a5601a38278a1467c2`;
- results `272a09e62c1b9b36eabc167cdafaa2845fbe9198`.

The $\tilde{d} = 0.066$ main run and the controls used the binary built from `17ea6be` (before `cargo fmt`, a formatting-only change).

```
taskset -c 4-7 python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Autoresearch/
    DipolarExcitonMultipolar.lean \
  --olean Autoresearch.DipolarExcitonMultipolar --json-out results/autoresearch/2026-10-25/H1/contribution/
    compile_attempt1.json
taskset -c 4-7 python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Autoresearch/
    DipolarExcitonMultipolarControls.lean \
  --json-out results/autoresearch/2026-10-25/H1/contribution/compile_controls.json
taskset -c 4-7 python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Autoresearch/
    DipolarExcitonStep1.lean \
  --olean Autoresearch.DipolarExcitonStep1 --json-out results/autoresearch/2026-10-25/H1/contribution/
    compile_step1.json
taskset -c 4-7 python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Autoresearch/
    DipolarExcitonStep1Controls.lean \
  --json-out results/autoresearch/2026-10-25/H1/contribution/compile_step1_controls.json
CARGO_TARGET_DIR=... cargo test --release -j 2 --offline --manifest-path <rusty-wt>/Cargo.toml -p qf-yukawa-
    crystal -- --test-threads=2
dipolar_crystal refs|gaps|controls <crate>;  dipolar_crystal main <crate> 0.066 8 2
DIPOLAR_THREADS=1 dipolar_crystal main <crate> 0.0209 2 0 0;  DIPOLAR_THREADS=1 dipolar_crystal main <crate>
     1 2 0 0
python -I scripts/dipolar_hf_download.py <hf-data>/dipolar_exciton colabfit/
    JARVIS_DFT_3D_12_12_2022@f4de0085...:co/co_0.parquet ...
python -I scripts/dipolar_tmd_extract.py ... ; python -I scripts/dipolar_physical_d.py ...
python -I scripts/dipolar_independent_check.py ... ; python -I scripts/dipolar_figures.py ...
python -I papers/ar_dipolar_exciton/ledger/build_ledger.py ; python -I papers/ar_dipolar_exciton/ledger/
    run_gate.py
python -I papers/ar_dipolar_exciton/gen_inputs.py ; pdflatex main.tex (twice)
```

Paper-stage reruns ([X, DX-X-0008]), by the same AI pipeline, not by a person: crate tests 28 passed, 0 failed, 1 ignored (the ignored test predates this work); Lean recompile of the controls file (imports the stage-1 oleans, so it re-reads axioms rather than re-elaborating the module): rc 0 in 755.4 s, axiom output identical to stage 1; Lean recompile of the Multipolar source without writing oleans (re-elaborates the new module against the prior-round olean): rc 0 in 660.7 s, axiom output identical to the stage-1 compile. Neither rerun separates producer from verifier.

References

[1] OpenAI, *Universal optimality of the triangular lattice*, OpenAI Math release, github.com/openai/math, commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb` (2026). AI-generated; not peer reviewed.

[2] M. Petrache, S. Serfaty, *Crystallization for Coulomb and Riesz interactions as a consequence of the Cohn–Kumar conjecture*, arXiv:1908.09714v3 (2019).

[3] J. Wang, Q. Shi, E.-M. Shih, L. Zhou, W. Wu, Y. Bai, D. A. Rhodes, K. Barmak, J. Hone, C. R. Dean, X.-Y. Zhu, *Diffusivity reveals three distinct phases of interlayer excitons in $MoSe_2/WSe_2$ heterobilayers*, arXiv:2001.03812v5.

[4] K. Choudhary et al., *The Joint Automated Repository for Various Integrated Simulations (JARVIS) for data-driven materials design*, arXiv:2007.01831v2; publication doi:10.1038/s41524-020-00440-1 as given in the dataset card (journal volume and page not verified by us).

[5] K. Choudhary et al., *JARVIS DFT 3D 12 12 2022*, ColabFit (2023), doi:10.60732/e9e65ccd; HuggingFace `colabfit/JARVIS_DFT_3D_12_12_2022`, revision `f4de0085edcf92885870c67459363d2c7080c90b`.

[6] X. Callens, *Riesz-energy, finiteness and every-density corollaries of the openai/math planar universal-optimality theorem, kernel-checked in Lean 4*, doi:10.5281/zenodo.23285631 (2026); D2 is its every-density module `formal_cert/Triangular/AnyDensity.lean`.

[7] X. Callens, *Planar universal optimality, tiered: kernel-checked consequences, a non-uniqueness theorem, and a CVODE N-body consistency check for the Yukawa interaction*, preprint v2, doi:10.5281/zenodo.23293429 (2026).

[8] X. Callens, *The Triangular Lattice and Universal Optimality* (textbook), doi:10.5281/zenodo.23295142 (2026).

This preprint applies the approach of [7, 8] (kernel-checked consequences of upstream for explicit interactions) to the bilayer dipolar-exciton interaction.